

Improving sentiment analysis in tourism through LLM-enhanced irony detection

Wenhui Liu, Likang Wu, Hongke Zhao¹ *

College of Management and Economics, Tianjin University, Tianjin, 300072, China
Laboratory of Computation and Analytics of Complex Management Systems (CACMS), China

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ABSTRACT

As user-generated content continues to proliferate across digital tourism platforms such as TripAdvisor, it has become a critical resource for evaluating service performance and tourist satisfaction within the tourism sector. Among these reviews, ironic ones often reveal underlying issues or dissatisfaction, reflecting more genuine and reliable human emotions to influence users' choices. However, conventional sentiment analysis methods often fail to recognize irony, which can express negative sentiments through a superficially positive tone. To address the challenge of accurately classifying sentiments in tourism reviews, particularly ironic ones that can mislead traditional sentiment analysis models, this study introduces an innovative sentiment analysis framework named TSI. Leveraging large language models (LLM) for data augmentation, our approach integrates transfer learning and domain-specific knowledge, enhancing detection accuracy through advanced natural language processing (NLP) techniques and deep learning technologies. Extensive experiments on travel-related datasets from Yelp and TripAdvisor demonstrate that TSI outperforms traditional models in both overall sentiment classification accuracy and the identification of negative sentiments, particularly in ironic reviews. This work advances theoretical understanding and practical implementation of sentiment analysis within the tourism sector, equipping businesses with refined tools for interpreting customer sentiments. A new dataset targeting ironic travel reviews is also released to support ongoing scholarly exploration. For better reproduction, the source code is available.¹

1. Introduction

The tourist experience is deeply influenced by emotional factors at multiple stages (Prayag et al., 2013), particularly before the journey begins, when emotions guide both the desire to travel and the selection of travel destinations (Goossens, 2000). During the trip, emotional intensity can fluctuate on a daily basis (Hosany et al., 2020). Moreover, emotional responses from tourists are key determinants in how they assess their travel experiences afterward, including their level of satisfaction (del Bosque & Martín, 2008), destination attachment (Yuksel et al., 2010), trust, commitment (Han & Jeong, 2013), and loyalty (Yoon & Uysal, 2005). Especially negative emotional experiences often lead to reduced levels of satisfaction and customer loyalty (Lee et al., 2008). As social media and online review sites have rapidly expanded, content created by users has emerged as a key resource for assessing service quality and tourist satisfaction (Polyzos et al., 2023). Users share their travel experiences, insights, and

comments on these platforms, providing valuable information for both tourists and tourism-related businesses (Xu et al., 2022). For instance, Xiaohongshu, a representative social platform from China, that exhibits outstanding performance in cultural and tourism user-generated content (UGC) marketing, has surpassed 312 million monthly active users.²

In recent years, the presence of ironic sentiment in online reviews has increased, primarily due to the implicit expression of consumer sentiment (Joshi et al., 2017). Irony is a common rhetorical device on social media platforms (Guan et al., 2025a), occurring in up to 8.5% of online reviews³ and is difficult to recognize by software tools (Papapicco & Mininni, 2020). Linguistic studies suggest that irony arises when an incongruity exists between the situational context and the literal meaning conveyed by a statement (Gerrig & Goldvarg, 2000;

* Corresponding author at: College of Management and Economics, Tianjin University, Tianjin, 300072, China.
E-mail address: hongke@tju.edu.cn (H. Zhao).

¹ <https://github.com/252820/Tourism-Model>

² <https://www.netinfluencer.com/xiaohongshu-social-media-investment-dst-global/>

³ <https://gupea.ub.gu.se/handle/2077/23031>

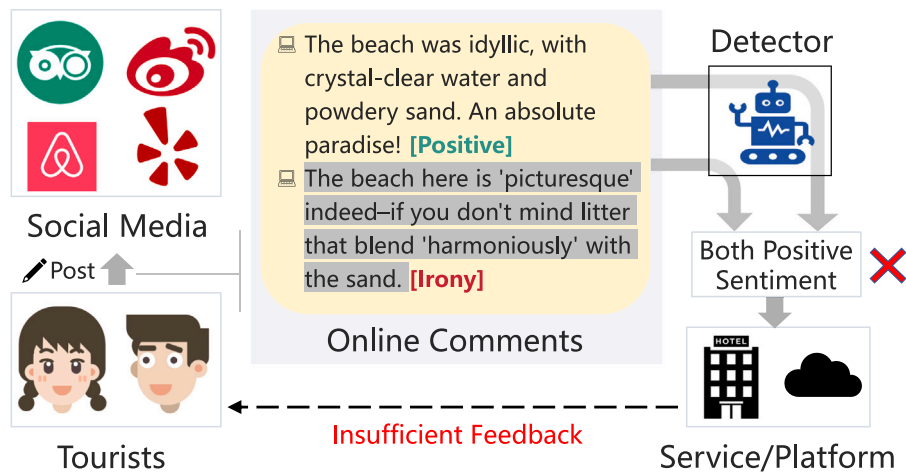


Fig. 1. Effect of ironic review.

Ivanko & Pexman, 2003; Joshi et al., 2015; Zhang et al., 2019). When applied to tourism review texts, we define irony as “verbal irony characterized by sentiment reversal, where the sentiment expressed contradicts the literal meaning. Specifically, the literal expression conveys a positive sentiment, while the intended meaning is negative, or vice versa”. Ironic elements can induce machines to focus on superficially positive descriptive words, thus making opposite sentiment judgments on the comments (Sarmiento et al., 2009). Moreover, compared with the large amount of monotonous and artificially generated content released by some platforms and merchants for marketing purposes, ironic sentences that reflect more genuine and delicate human emotions (Bajerski, 2016) are more reliable and worthy of reference for users. Given that 70.9% of travelers acknowledge the influence of online reputation on their travel decisions,⁴ it is essential to examine this factor in tourism contexts such as online travel planning. Notably, when negative reviews exhibit heightened emotional intensity and authenticity, they tend to significantly undermine trust in the destination (Lujun et al., 2021).

Given an intuitive case, in Fig. 1, a positive review praises a beach as “idyllic” with “crystal-clear water” and “powdery sand”, emphasizing its appeal. In contrast, an ironic review describes the same beach as “picturesque”, pointing out the presence of litter that “harmoniously” blends with the sand, humorously highlighting a negative aspect. However, sentiment analysis detector classifies both genuinely positive and ironic reviews as “positive sentiment”, failing to distinguish between true positivity and irony. This results in misleading outcomes, hindering platforms and service providers — such as hotels and tourist destinations — from obtaining accurate feedback and effectively identifying and addressing the issues raised by users. For instance, if a review like “This restaurant is truly unique, it can turn a simple salad into a disaster!” is misclassified as expressing positive sentiment, it can lead to a misjudgment of tourist feedback by service providers. Such misclassification obscures underlying issues, rendering feedback analysis ineffective.

Modern travel platforms such as TripAdvisor and Yelp allow users to share their experiences through comments and ratings, offering important attitudinal and emotional feedback to service providers (Tsai et al., 2020). Nevertheless, language often carries subtle emotional cues that are not fully captured by conventional rating systems (Geetha et al., 2017). Additionally, some platforms and social media sites do not offer rating features, thereby restricting users’ ability to efficiently interpret the sentiment of reviews. For instance, platforms like Instagram and Xiaohongshu lack integrated rating tools, which pose challenges for

both users and platforms in deriving valuable insights, particularly when attempting to detect irony or deeper emotional expressions.

Text sentiment analysis is a crucial task for promoting numerous tourism applications (Sharma, Adhikary, & Borah, 2020). It involves detecting and categorizing sentiments embedded in user-generated content related to tourism by leveraging advanced algorithms (Jim et al., 2024). The purpose is to assess whether the author expresses a positive, negative, or neutral stance regarding a given topic, product, or service (Wankhade et al., 2022). Due to the openness of tourism review communication, consumers tend to approach it with an emotional attitude and often express negative sentiment in their reviews (Goldin et al., 2008). Conducting sentiment analysis on these reviews can improve the sales performance of enterprises, especially in the tourism industry (Rialp-Criado et al., 2018). However, traditional sentiment analysis tools, while effective in broad strokes of positive or negative sentiment detection, fail when encountering the complexities of human emotions and linguistic expressions such as irony (Balamurali et al., 2024; Hu & Liu, 2012). Current algorithms primarily focus on the transformation of general language models, with insufficient capability to recognize subtle implied meanings and emotions (Chakravarthi et al., 2021). Additionally, they lack adequate integration and modeling of domain-specific knowledge, particularly in the context of common tourism-related information.

To tackle this challenge, we introduce a novel Text Sentiment classification model that incorporates Irony knowledge, TSI, specifically designed to accurately classify the sentiment of ordinary and ironic travel reviews. By integrating large language model (LLM)-based data augmentation, domain-specific irony patterns, and context-aware sentiment modeling, the proposed TSI framework addresses key limitations in prior work and enables more accurate detection of sentiment in tourism reviews. The approach is designed based on the following three views.

- V1: Addressing the ironic reviews sparsity issue can improve the algorithm’s ability to perceive irony.
- V2: The ironic knowledge available in the public domain can enhance the model’s ability to detect irony.
- V3: Leveraging knowledge from the tourism field can aid in designing a model tailored for analyzing travel reviews.

We will validate these viewpoints through experiments. The approach we proposed combines semantic analysis with context-aware techniques to detect the implicit sentiment cues embedded in irony, thereby avoiding surface-level misinterpretations. Specifically, this method consists of three main stages. First, due to data scarcity, we use a large language model (LLM) for data augmentation, generating ironic travel

⁴ <https://www.mara-solutions.com/post/online-review-statistics-every-hotel-needs-to-know-in-2023#>

reviews to mitigate the issue of data imbalance. Second, given the pattern similarities commonly found in ironic expressions, cross-domain transfer learning offers a promising solution to the scarcity of irony-related knowledge within the tourism sector (Celik et al., 2020; Guan et al., 2025a; Guo et al., 2021; Kadli & Vidyavathi, 2023). Therefore, we combine ironic knowledge transfer learning with keyword weight adjustment tailored to the tourism domain, enhancing the model's understanding of ironic expressions and improving the accuracy of tourism review classification. Finally, we evaluate the model's accuracy and practicality through various experiments and a case study. Compared to existing approaches, our model shows significant improvements in accuracy while maintaining generalizability and scalability across diverse datasets. As far as we are aware, this is the first study to utilize LLM for generating and classifying ironic content in travel reviews. Our findings validate the potential of LLMs for data augmentation and contribute to broadening the scope of sentiment analysis within the tourism domain. A detailed description of our method can be found in Section 3.

This study contributes to three aspects. First, this study tackles the challenge of identifying sentiment in ironic tourism reviews by proposing a robust approach that advances sentiment analysis in this domain. By resolving the subtle complexities of irony, our approach provides a more reliable framework for understanding user feedback in the context of tourism. Second, our work extends beyond theoretical contributions to have practical implications in areas such as social media review analysis, customer feedback interpretation, and enhancing user experience. Our work not only deepens the understanding of review content by going beyond simple numerical ratings but also uncovers the sentiment complexity embedded in user opinions. By accurately discerning the true sentiment behind ironic reviews, our method helps businesses and platforms gain deeper insights into customer experiences. This enables them to better identify areas for improvement, tailor their services more effectively, and ultimately improve customer satisfaction and engagement. Third, to support further research in this area, we have made a new dataset publicly available, specifically designed to examine ironic comments in tourism reviews. The dataset offers a meaningful foundation for scholars seeking to investigate how irony influences consumer perception and decision-making processes within the tourism sector.

The structure of this paper is as follows: Section 2 outlines relevant literature in the field; Section 3 elaborates on the methodology adopted in this study. Section 4 reports the experimental design and corresponding results, while Section 5 concludes with a summary of insights and their broader implications.

2. Literature review

2.1. Sentiment analysis in tourism

UGC from online travel platforms has become a primary data source for tourism research, capturing travelers' experiences, sentiments, and decision-making processes (Wang, Huang, & Liu-Lastres, 2022). Accurately extracting and interpreting emotions from tourism-related texts is essential for understanding tourist behavior and promoting the sustainable development of the tourism industry (Alaei et al., 2019; Feldman, 2013).

Recent studies have demonstrated that sentiment analysis not only supports destination image construction and brand promotion but also significantly influences traveler decision-making and satisfaction (Hosany et al., 2020; Ladhari, 2009; Maital, 1999; Neidhardt et al., 2017). For instance, Li, Gao, and Song (2023) integrated sentiment analysis of user reviews to enhance forecasting models for tourism demand, while (Liu et al., 2019) applied sentiment modeling to evaluate perceptions of Australian destinations among Chinese tourists. Other studies have used sentiment signals to assess destination carrying

capacities (Tokarchuk et al., 2022) and to predict business outcomes such as restaurant survival (Li, Yu, et al., 2023).

The utility of sentiment analysis extends to uncovering customer preferences regarding hotels (Bian et al., 2022; Tsai et al., 2020), supporting the development of recommendation systems (Ray et al., 2021), and forecasting financial performance in the tourism sector (Polyzos et al., 2024). Empirical evidence consistently links review sentiment to customer ratings and satisfaction (Geetha et al., 2017; Sharma, Park, & Nicolau, 2020), underlining its practical importance for strategic decision-making (Jain et al., 2022).

However, most existing studies have focused on aggregate sentiment signals while often overlooking nuanced linguistic expressions — such as irony — which may compromise classification accuracy. The limited attention to such subtleties constrains the operational effectiveness of current approaches in complex real-world scenarios.

2.2. Methodology of sentiment analysis

The evolution of sentiment analysis methods in tourism has closely followed broader developments in natural language processing. Early research relied on sentiment lexicons and basic machine learning classifiers (Pang & Lee, 2008; Turney & Littman, 2003). Algorithms such as Naïve Bayes (Su et al., 2011), LightGBM (Ke et al., 2017), and SVM (Pu et al., 2019) achieved initial success but were limited by their reliance on labor-intensive feature engineering (An & Moon, 2022).

With the emergence of deep learning architectures — including CNN (Kim, 2014), BERT (Sousa et al., 2019), Transformer (Naseem et al., 2020), TextCNN (Wang, Wang, et al., 2022), and Bi-LSTM (Cheng & Chen, 2021) — the accuracy of sentiment classification has improved significantly, particularly for longer and more complex reviews (Li, Li, et al., 2023; Paolanti et al., 2021). These models enable automated feature extraction and robust context modeling; however, their performance still degrades in the presence of sophisticated language phenomena such as irony (Balamurali et al., 2024).

Transfer learning has emerged as a promising strategy to address domain heterogeneity and improve model generalization (Ali et al., 2023; Celik et al., 2020; Guo et al., 2021; Liang et al., 2021). Recent work applies transfer learning not only within sentiment analysis tasks but also across related tasks such as irony detection (Guan et al., 2025b; Zhang et al., 2019). Nevertheless, the integration of domain-specific knowledge and explicit handling of irony remains underexplored in tourism text analytics.

In summary, despite significant technological advances, existing studies often lack domain-specific adaptation, leading to suboptimal performance in tourism-related sentiment analysis. To address this limitation, this study proposes a novel approach that integrates transfer learning with domain-specific knowledge, aiming to improve both the accuracy of sentiment polarity classification and the depth of contextual understanding in tourism text analysis.

3. Methodology

This section will introduce the irony-aware sentiment recognition model that we proposed, along with the overall framework for performing our task.

3.1. TSI model

The TSI model consists of four key modules: Tourism Domain Enhancement Module, Textual Attention Computing Module, Irony-Aware Transfer Learning Module, and Sentiment Classification Module. This model effectively analyzes tourism review data by integrating public domain knowledge, BERT's pre-trained knowledge, and the strengths of the self-attention architecture. By doing so, it can thoroughly capture the semantic information across the entire review. Fig. 2 provides an illustration of the TSI model's structure.

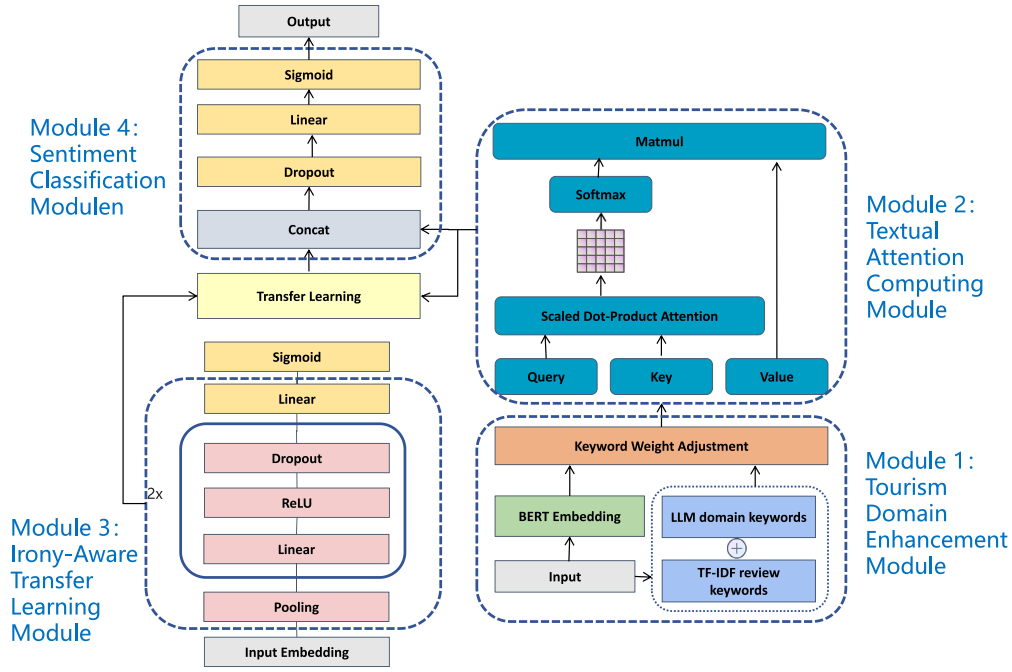


Fig. 2. Architecture of the proposed TSI model.

(1) Module 1 : Tourism Domain Enhancement Module

To design a model tailored for analyzing travel reviews, this module integrates BERT embeddings with domain-specific knowledge from the tourism field (i.e., a set of relevant keywords) to construct features. First, the BERT model is used to convert the input review text into an embedding vector.

$$\mathbf{e}(\mathbf{x}) = \text{BERT}(\mathbf{x}), \quad (1)$$

where $\mathbf{x}$ represents the input review text, and $\mathbf{e}(\mathbf{x})$ is the embedding vector obtained by processing the input review text through the BERT model. $\text{BERT}(\cdot)$ denotes the encoding process carried out by the BERT model. BERT (Devlin et al., 2019), a pre-trained model, excels in capturing deep semantic relationships within text. Its primary strength lies in its bidirectional Transformer architecture, which enables enhanced text comprehension by effectively capturing contextual information. Then, to further enhance knowledge in the tourism domain, we adjust the weight of each keyword in the domain keyword list, increasing its embedding weight to $(1 + p)$ times its original value. Here, p is a hyperparameter that controls the weight increase ratio, which lies between 0 and 1.

$$\mathbf{E} = \mathbf{e}(\mathbf{x}) + p \cdot \mathbf{e}(\mathbf{x}) \odot \mathbf{J}, \quad (2)$$

where $\mathbf{E}$ represents the final embedding vector after weight adjustment, and $\odot$ denotes the element-wise multiplication (Hadamard product). The indicator function $\mathbf{J}$ indicates whether the i th dimension of the embedding corresponds to a domain keyword. If $J_i = 1$, the corresponding weight is amplified; if $J_i = 0$, the weight remains unchanged.

The domain keyword list consists of two components: one derived from large language models (i.e., LLM domain keywords) and the other from the review texts (i.e., TF-IDF review keywords). The LLM domain keywords are generated by GPT-4 using prompt engineering, which helps guide the model to produce relevant tourism-related keywords. GPT-4, a powerful large model with extensive public domain knowledge, is capable of capturing essential knowledge in the tourism domain. Prompt engineering provides a clear direction on the type of content to generate. The TF-IDF review keywords consist of the top 100 high-frequency terms extracted from the review texts in the dataset using the TF-IDF method. This technique, commonly applied in information retrieval and text mining, is designed to identify key

terms for tokenization (Chen et al., 2016; Kim et al., 2019). TF-IDF assesses a word's importance within a document by evaluating both its frequency in the document and its rarity across the entire corpus. A higher TF-IDF score reflects a word's greater significance. This approach effectively emphasizes crucial vocabulary while minimizing storage and computational demands (Tian & Zhang, 2023).

In summary, this module utilizes BERT embeddings to generate rich, contextual semantic representations of the review text, capturing both the overall meaning and intricate relationships between words. At the same time, the keyword weight adjustment process amplifies the significance of key features directly related to the tourism domain. This method integrates both domain-specific knowledge and review-specific terminology, ensuring the model prioritizes the most relevant terms. The combined focus on contextual comprehension and domain relevance strengthens the model's capability to accurately identify sentiment in tourism reviews, thereby improving the overall sentiment analysis process.

(2) Module 2 : Textual Attention Computing Module

In this module, we employ the self-attention mechanism to calculate word-level attention for the review text. The process begins by applying three distinct linear transformations to the embedding vectors of each word to generate the query Q , key K , and value V representations, respectively. Next, attention scores are computed by calculating the dot product between Q and K , scaled by $\sqrt{d_k}$, where d_k is the dimension of the key vector. The scores are then passed through the softmax function to obtain the normalized attention weights A . Finally, the context-aware representation C for each word is obtained by multiplying the attention weights matrix A with the values matrix V . This procedure can be expressed as follows.

$$\begin{aligned} Q &= \mathbf{E}\mathbf{W}^Q, & K &= \mathbf{E}\mathbf{W}^K, & V &= \mathbf{E}\mathbf{W}^V, \\ A &= \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right), \\ C &= AV, \end{aligned} \quad (3)$$

where $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V$ represent learnable parameter matrices.

The self-attention mechanism assigns distinct weights to each word in the input sequence, allowing the model to capture relationships between words regardless of their positions in the sentence (Vaswani

et al., 2017). The key concept of self-attention is to compute the attention weight for each word, reflecting its importance, and then aggregate the information of the words based on these weights. This allows the model to focus on significant features while suppressing less relevant ones, effectively enhancing its ability to capture nuanced contextual relationships. By incorporating self-attention, the model becomes more sensitive to the sentiment nuances in the text, better capturing the subtle shifts in sentiment tendencies throughout the review. This results in a richer understanding of textual sentiment content, ultimately improving sentiment classification performance.

(3) Module 3 : Irony-Aware Transfer Learning Module

The lack of sufficient contextual knowledge and specific domain information complicates the accurate detection of sentiment in ironic statements. Utilizing external sentiment knowledge is a highly effective strategy for improving irony detection (Zhang et al., 2019). To capitalize on the extensive ironic knowledge available in public resources and enhance the model's ability to identify and interpret irony, we apply transfer learning in this module to improve sentiment representation. Specifically, we adapt a Deep Neural Network (DNN) model, previously trained on an irony dataset, to a tourism review dataset. Transfer learning, a machine learning technique, boosts model performance by transferring knowledge gained from one task to a related task (Du et al., 2013). The core idea behind transfer learning is to transfer annotated data or knowledge structures from the source domain to the target domain, thereby improving the learning effectiveness for the target task (Pan & Yang, 2010). In our model, the source domain involves the task of irony recognition using an irony dataset, whereas the target domain focuses on identifying the sentiment of ironic reviews within tourism contexts. The transfer learning process involves transferring model parameters between the source and target domains to improve performance in the target domain. The procedure is described in detail as follows.

Initially, a DNN model is trained on the source domain dataset, D_s . A binary cross-entropy loss function, L_s , is employed to optimize the model and derive the source domain model parameters, θ_s . The training process can be expressed mathematically as:

$$\theta_s = \arg \min_{\theta} L_s(f(D_s; \theta)). \quad (4)$$

Here, $f(\cdot; \theta)$ represents the model function of DNN and θ denotes the model parameters. The source domain dataset D_s is a publicly available irony dataset. The DNN model comprises an embedding layer, a pooling layer, three fully connected layers, two ReLU activations, two Dropout layers, and a Sigmoid activation. Input text data is embedded into a vector space using GloVe embeddings. The loss function adopts binary cross entropy loss $\mathcal{L}_s$, given by:

$$\mathcal{L}_s = -\frac{1}{N} \sum_{i=1}^N [(1 - z_i) \cdot \log(1 - p(z_i)) + z_i \cdot \log(p(z_i))], \quad (5)$$

where $p(z_i)$ represents the predicted probability for the i th item, z_i represents the true label for the i th item, and N represents the number of items. The model is trained using the Adam optimizer, which is a variant of stochastic gradient descent (SGD) that incorporates adaptive moment estimation (Kingma, 2014).

We utilize the learned parameters to initialize the model for the target domain after training the model on the source domain. The dataset for the target domain is comprised of tourism reviews. The source domain parameters, denoted as θ_s , are used to initialize the target domain model. Subsequently, the target model undergoes fine-tuning with the target domain dataset. The fine-tuning procedure can be expressed mathematically as follows:

$$\begin{aligned} \mathbf{T} &= f(\mathbf{C}; \theta_t), \\ \theta_t &= \theta_s, \end{aligned} \quad (6)$$

where $f(\cdot)$ is forward propagation function of the DNN model, and θ_t denotes the model parameters for the target domain. By transferring the learned ironic knowledge, the model can apply the acquired understanding of irony to recognize ironic sentiment in tourism reviews.

Through the use of transfer learning, the model can effectively transfer ironic knowledge from the source domain, thereby improving the detection of ironic sentiment in the target domain—tourism reviews. This approach not only improves the performance of the model in sentiment classification but also allows for more accurate detection of irony, which is often challenging to identify using traditional sentiment analysis techniques.

(4) Module 4 : Sentiment Classification Module

In the sentiment classification module, the feature representations of module 2 and module 3 are first combined using a concatenation layer. The concatenated features are expressed as follows:

$$\mathbf{H} = \text{Concat}(\mathbf{C}; \mathbf{T}). \quad (7)$$

Following this, a Dropout layer is applied to mitigate the risk of overfitting by randomly disabling connections in the network, which helps enhance the generalization ability of the model. Afterward, the concatenated features are fed into a linear layer to make the final sentiment classification. The output from this layer is then passed through the Sigmoid activation function, which normalizes the result to a range between 0 and 1. This final output is represented as:

$$\begin{aligned} \mathbf{H}_d &= \text{Dropout}(\mathbf{H}), \\ \mathbf{Z} &= \mathbf{H}_d \mathbf{W}_l + \mathbf{b}_l, \\ \hat{t} &= \sigma(\mathbf{Z}) = \frac{1}{1 + e^{-\mathbf{Z}}}. \end{aligned} \quad (8)$$

Here, $\mathbf{W}_l$ denotes the weight matrix of the linear layer, $\mathbf{b}_l$ represents the bias term, and $\hat{y}$ represents the final sentiment classification result. The value of $\hat{y}$ close to 1 indicates a higher likelihood of positive sentiment, while the value close to 0 signifies negative sentiment. The final output is classified into specific sentiment categories, such as positive (1) or negative (0).

(5) Model training

For each tourism review, the true label t_j and the predicted sentiment result $\hat{t}_j$ from the model are computed. The performance of the model is assessed using the binary cross-entropy loss. The objective is to minimize this loss by optimizing the model parameters. This can be expressed mathematically as:

$$\min_{\phi} J(\phi) = -\frac{1}{M} \sum_{j=1}^M [t_j \log(\hat{t}_j) + (1 - t_j) \log(1 - \hat{t}_j)]. \quad (9)$$

$J(\phi)$ represents the loss function, which depends on model parameters ϕ . M denotes the number of reviews. We perform model optimization using the Adam optimizer, a widely used adaptive learning rate method that efficiently updates the model parameters.

3.2. The proposed overall framework

This section presents the overall framework for conducting the task. The procedure is divided into three main stages: data preparation, data processing and model development, and model evaluation. The workflow of the proposed overall framework is depicted in Fig. 3, providing a clear visualization of the model components and their interactions.

Step 1 : Data preparation

The initial step in the proposed overall framework consists of preparing four datasets, which come from two primary sources: the public dataset and the LLM Irony Dataset.⁵ Sarcasm Corpus V2⁶ is a

⁵ <https://github.com/252820/Tourism-Model/blob/master/data/irony.csv>

⁶ https://github.com/soraby/sarcasm2/tree/main/sarcasm_v2

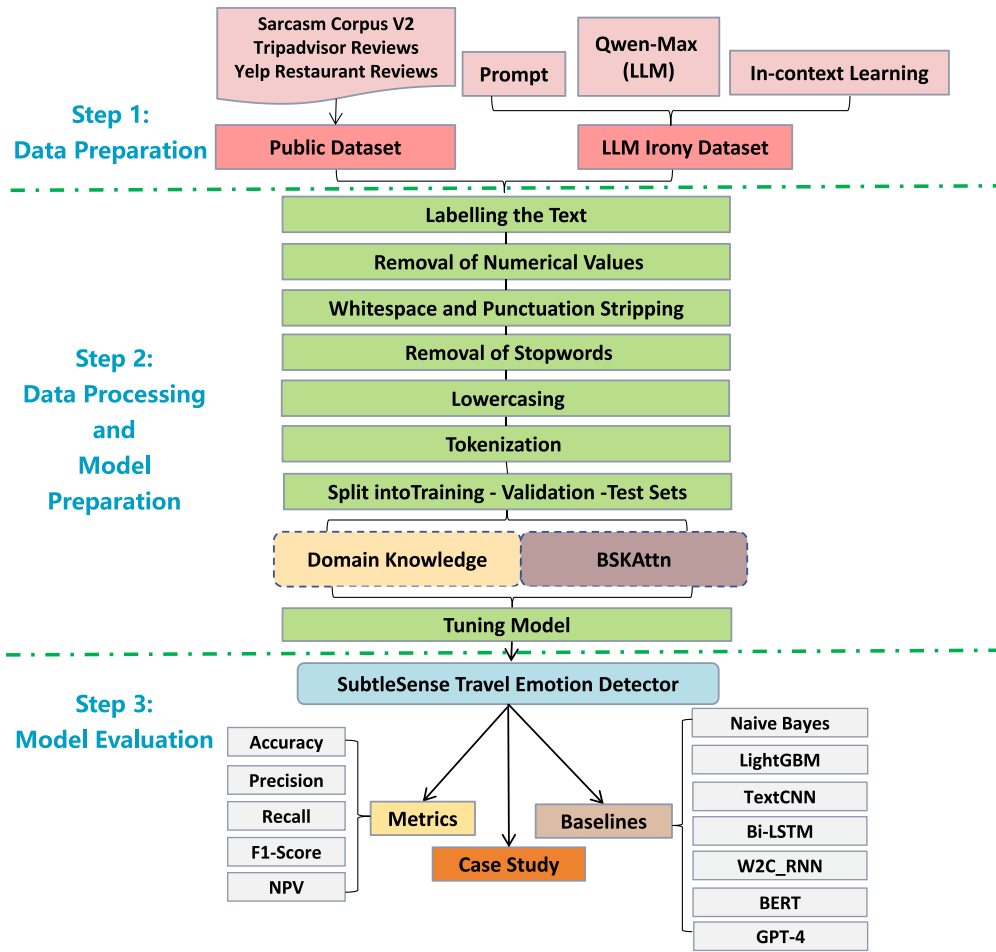


Fig. 3. Overall framework.

commonly used dataset for irony detection. It is utilized as the source domain dataset for performing DNN transfer learning in Module 2. Tripadvisor Reviews⁷ and Yelp Restaurant Reviews⁸ are commonly utilized datasets for sentiment analysis within the tourism sector (Ali et al., 2022; Xiang et al., 2017). [LLM Irony Dataset is a dataset generated using LLM, aimed at alleviating the sparsity issue of ironic reviews]. A comprehensive explanation of the dataset synthesis process is outlined below. All datasets are publicly accessible and can be obtained via the links provided in the footnotes.

Implicit sentiment conveyed through irony is challenging for traditional models, and ironic reviews are rare in available tourism datasets, limiting sentiment analysis performance in this domain. To address this, we used an LLM for data augmentation, generating additional ironic reviews with negative sentiment to expand our dataset. Specifically, we first manually selected 300 ironic sentences from 3000 negative-sentiment reviews sourced from Tripadvisor and Yelp, forming a seed sample pool D_p . Using in-context learning, we repeatedly prompted the Qwen-Max LLM: in each iteration, two random seed samples were inserted into a fixed prompt template (see Fig. 4) to generate a new ironic review. This process was repeated 5000 times, producing 5000 candidate samples, mathematically expressed as:

$$S_g = \text{LLM}(\text{prompt}_g(S_1, S_2)), \quad (10)$$

where $S_1, S_2 \in D_p$. After deduplication and manual quality screening, 3000 high-quality ironic reviews were retained as our final LLM Irony

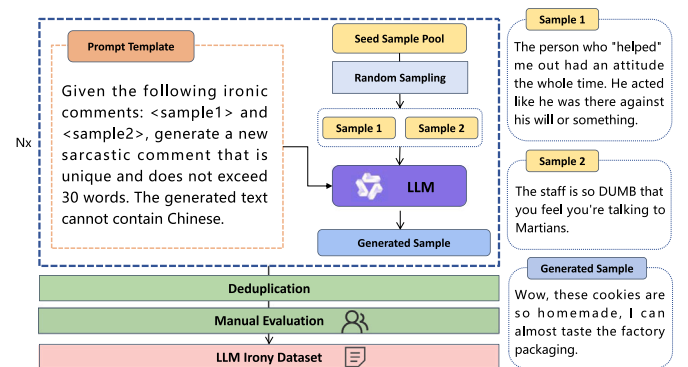


Fig. 4. Data augmentation process.

Dataset, all labeled with negative sentiment. This approach significantly broadens the diversity of ironic expressions in our dataset and enhances the model's ability to recognize implicit sentiment in tourism reviews.

Step 2: Data processing and model preparation

Since the original datasets lack sentiment polarity labels, we manually labeled the review texts for our experiments. Sentiment polarity was treated as a binary classification task, where "0" indicates negative sentiment and "1" indicates positive sentiment. Specifically, reviews rated 4 or 5 were labeled as positive (1), while those rated between 1 and 3 were categorized as negative (0). In most cases, a 3-point rating

⁷ <https://www.kaggle.com/datasets/arnabchaki/tripadvisor-reviews-2023>

⁸ <https://www.kaggle.com/datasets/farukalam/yelp-restaurant-reviews>

indicates that the user perceives the service as merely meeting basic standards, failing to generate significant joy or satisfaction (Ali et al., 2022; Li et al., 2024; Xiang et al., 2017). Such reviews often suggest that the service was mediocre, failing to leave a strong positive impression or enhance customer satisfaction. To encourage tourism service providers to deliver better experiences, increase customer satisfaction, and improve the likelihood of repeat visits, we have conservatively included reviews with a 3-point rating in the “negative” category. This binary labeling approach was chosen to simplify the sentiment analysis process and facilitate more efficient model training. By adopting this clear-cut classification method, we aimed to streamline the analysis and improve the model’s capability to accurately distinguish between positive and negative sentiments in tourism reviews.

Following the labeling process, several data cleaning procedures were carried out to ensure the dataset’s quality and consistency. First, all numerical values were removed from the text to prevent any unintended influence on sentiment classification. Next, excess whitespace and unnecessary punctuation were eliminated to streamline the text and improve processing efficiency. Common stopwords that do not contribute significant meaning for sentiment analysis were filtered out. All text was transformed to lowercase to standardize the data and reduce token variability. Each review was tokenized to facilitate more efficient model processing and analysis. These preprocessing steps were crucial for standardizing the data, minimizing noise, and improving the efficiency and accuracy of subsequent tasks. After preprocessing, the dataset was divided into training, validation, and test sets.

By following these rigorous preprocessing steps, we ensured that the dataset was well-prepared for training high-performing sentiment analysis models, thereby enhancing their ability to accurately classify the sentiment of tourism reviews. Then we integrated domain knowledge into the proposed model and used the prepared data to train and fine-tune it.

Step 3: Model evaluation

To assess the effectiveness of our approach, we utilized five commonly recognized evaluation metrics: accuracy, precision, recall, F1-score, and Negative Predictive Value (NPV). These metrics are commonly employed for evaluating machine learning models (Choi et al., 2020). These metrics are defined as follows:

$$Acc = \frac{TP + TN}{TP + FP + TN + FN}, \quad (11)$$

$$Pre = \frac{TP}{TP + FP}, \quad (12)$$

$$Rec = \frac{TP}{TP + FN}, \quad (13)$$

$$F1 = \frac{2 \times Pre \times Rec}{Pre + Rec}, \quad (14)$$

$$NPV = \frac{TN}{TN + FN}. \quad (15)$$

In these equations, TP represents the number of positive reviews in the gold-standard dataset that are correctly identified as positive, while TN denotes the number of negative reviews accurately classified as negative. Conversely, FN refers to positive reviews mistakenly classified as negative, and FP signifies the reviews predicted as positive that are actually negative in the gold-standard dataset.

Higher values for accuracy, precision, recall, F1-score, and NPV generally indicate better overall model performance (Bian et al., 2022).

We selected Naive Bayes, LightGBM, TextCNN, Bi-LSTM, W2C_RNN, BERT, and GPT-4 as our baseline models. Each of these models serves as a benchmark for evaluating the performance and efficacy of the TSI model. They cover a wide range of sentiment analysis techniques, from traditional machine learning methods like Naive Bayes and LightGBM to more sophisticated deep learning models such as TextCNN, Bi-LSTM, W2C_RNN, and BERT. This diverse set of baseline models provides a comprehensive evaluation framework, allowing us to thoroughly examine the performance and effectiveness of our proposed method across various evaluation metrics. By comparing our model against these varied approaches, we can better understand its strengths and weaknesses in different contexts.

4. Experiments

The absence of large-scale, high-quality datasets hinders the model’s ability to accurately detect sentiment (Guan et al., 2025a; Zhang et al., 2019). Transferring knowledge from one domain to another can lead to improved accuracy and outcomes (Celik et al., 2020). Utilizing domain-related information can enhance model performance (He et al., 2025). Based on these theoretical foundations, the following research questions are presented for the experimental section.

- **RQ1.** How can LLM be effectively utilized to alleviate the issue of ironic reviews sparsity?
- **RQ2.** How does the TSI model enhance sentiment analysis over baselines?
- **RQ3.** What is the effect of utilizing ironic knowledge available in the public domain?
- **RQ4.** What is the effect of utilizing knowledge from the tourism field?

These four questions correspond to the three Views mentioned in the introduction, i.e., they need to verify the motivation of our model design. To address these four questions, this section details the experimental setup, presents the results of comparison experiments, conducts an ablation study, and includes a case study to showcase the practical effectiveness of the TSI model.

4.1. Experimental setup

Machine learning models were implemented by constructing classifiers with the Scikit-Learn library in Python (Shoval et al., 2011), while deep learning models were built using the PyTorch framework, known for its adaptability and efficiency in handling large-scale deep learning tasks. All experiments were conducted on an Nvidia 4090 GPU with 24 GB of VRAM, ensuring fast processing and efficient handling of large datasets and model computations.

For model comparison, we evaluated the following seven baseline models:

- **Naive Bayes.** This probabilistic classifier, grounded in Bayes’ theorem, is commonly applied in sentiment analysis. For example, Al-muhaya et al. (2024) used polynomial Naive Bayes for sentiment analysis of Arabic tweets on Twitter, while (Charitha et al., 2023) employed it for sentiment classification of IMDB movie reviews.
- **LightGBM.** As an efficient Gradient Boosting Decision Tree (GBDT) framework, LightGBM is frequently used for sentiment analysis tasks. Khatun and Rabeya (2022) applied LightGBM to analyze Bangla-language book reviews, categorizing them as positive or negative.
- **TextCNN.** A deep learning model based on Convolutional Neural Networks, TextCNN, has shown effectiveness in text classification tasks (Kim, 2014). For example, Wang, Wang, et al. (2022) combined Word2Vec embeddings with TextCNN to perform sentiment classification on Weibo comments.
- **Bi-LSTM.** This network architecture improves upon the traditional Long Short-Term Memory (LSTM) model by processing data in both forward and backward directions, merging their outputs without altering the internal structure of LSTM (Gao et al., 2024). Zhang and Guo (2021) applied Bi-LSTM for sentiment analysis on product reviews from platforms like Taobao and JD.COM.
- **W2C_RNN.** This method combines Word2Vec embeddings with a Recurrent Neural Network (RNN), offering a novel approach particularly suited for travel contexts. Ying and Yufan (2023) used W2C_RNN for sentiment analysis of travel reviews to support destination recommendations.
- **BERT.** As a pre-trained NLP model, BERT (Devlin et al., 2019) has demonstrated outstanding performance across numerous NLP tasks. We included BERT as a baseline to determine whether performance improvements result from BERT’s capabilities or from the proposed methodology.

Table 1
Experiment parameter settings.

Model	Embedding dimension	Dropout	Epochs	Optimizer	Weight decay	Learning rate	Batch size
DNN	400	0.5	50	Adam	1e-5	1e-4	128
TSI	768	0.5	50	Adam	1e-4	1e-4	128

- **GPT-4.** Developed by OpenAI, GPT-4 is a large language model that utilizes a Mixture of Experts (MoE) architecture comprising approximately 1.76 trillion parameters. During a typical inference, around 220 billion of these parameters are activated. The model excels at tasks involving text processing, complex semantic reasoning, and long-context understanding.

To ensure reliable and reproducible classification performance across all models, the following systematic procedure was adopted:

- **Dataset Split.** The dataset was divided into three distinct subsets: 60% of the samples were designated for training, 20% for validation, and the remaining 20% for testing. This conventional division guarantees that the model is assessed on unseen data, offering a reliable estimate of its generalization ability.

- **Optimizer and Hyperparameter Tuning.** For optimization, the Adam optimizer was chosen due to its efficiency and adaptive learning rate, which helps improve convergence in complex models. Hyperparameters, including the learning rate, weight decay, and keyword weight parameter p , were tuned using grid search. This exhaustive search method tested different combinations of hyperparameter values to identify the optimal configuration for each model. The optimal parameter combinations and values that yielded the best performance are documented in the source code, ensuring transparency and reproducibility of the results.

- **Keyword Extraction.** In the keyword weight adjustment process, we utilized LLM to generate a total of 26 keywords, aimed at enhancing the quality of feature representation for the sentiment analysis task. In addition, we extracted another 100 high-frequency keywords using the traditional TF-IDF method.

- **Early Stopping.** To avoid overfitting and guarantee that the model does not memorize the training data, we adopted an early stopping strategy. This approach tracks the model's performance on the validation set throughout training. If performance stagnates without significant improvement for five consecutive epochs, training is stopped. This approach helps conserve computational resources and prevents the model from learning noise in the data.

The experimental parameters are set as shown in Table 1.

To solve RQ1, we used both GPT and Qwen-Max in the data augmentation process, combined with carefully designed prompts, to generate ironic reviews. Specifically, GPT was instructed to generate multiple reviews in a single request based on the provided prompt. Conversely, Qwen-Max generated multiple reviews through several API calls, incorporating random sampling into the prompt design. These two approaches resulted in noticeable differences in the outcomes of the generated reviews.

As illustrated in Fig. 5, compared to GPT's method of direct generation, Qwen's strategy using random sampling produced significantly more effective reviews. Fig. 5 demonstrates the authenticity and attractiveness of Qwen's generated content, as well as its advantages in creating engaging and realistic ironic comments. As observed, Qwen's approach stands out for generating nuanced, diverse, and contextually relevant content, frequently incorporating subtle details that enhance the engagement and realism of the reviews. In contrast, GPT's direct outputs tend to be repetitive, lacking the contextual richness and diversity needed to produce compelling irony. This comparison highlights the clear advantages of adding randomness and variability when augmenting samples with LLM.

4.2. Model evaluation

To further validate the contributions of each module in the TSI model, we conducted a stepwise validation and an ablation study using Yelp Restaurant Reviews and TripAdvisor Reviews.

(1) Stepwise validation

To validate the model in a stepwise fashion, we outline and evaluate three configurations. Detailed below are the specifics of each configuration, which will guide our phased assessment of the model's effectiveness.

- **I.** The model incorporates Module 1 and Module 4.
- **II.** The model incorporates Module 1, Module 2, and Module 4.
- **III.** The model incorporates Module 1, Module 2, Module 3, and Module 4.

Fig. 6 presents the results of the stepwise validation across the two datasets, demonstrating the effectiveness of each module.

(2) Ablation study

The ablation study provides answers to RQ3 and RQ4. The aim of this study was to evaluate the impact of the proposed modules — specifically Module 1 (the tourism domain enhancement module) and Module 3 (the irony-aware transfer learning module) — on overall performance. Below, we outline the three configurations evaluated:

- **TSI-KT.** The full model without the irony-aware transfer learning module and the tourism domain enhancement module.
- **TSI-T.** The full model without the irony-aware transfer learning module.
- **TSI.** The complete model, including all modules.

The outcomes of the ablation study for both datasets are presented in Fig. 7.

Fig. 7(a) illustrates the performance metrics for the Yelp Restaurant Reviews across the three model configurations. The results demonstrate that the TSI model consistently delivers the best performance across most metrics, clearly indicating the advantage of integrating both the irony-aware transfer learning module and the tourism domain enhancement module. The TSI model achieves superior accuracy and precision, suggesting that the inclusion of the irony-aware transfer learning module helps the model capture implicit irony patterns within the dataset. The TSI-T model shows a relative improvement in recall. This indicates that the tourism domain enhancement module positively impacts the model's ability to identify tourism-related features.

Similarly, Fig. 7(b) presents the performance metrics for TripAdvisor Reviews. The trends observed are almost consistent with those from the Yelp Restaurant Reviews, with the TSI model outperforming both TSI-T and TSI-KT in accuracy, precision, and F1-score. The TSI model continues to show superior performance, confirming its robustness across different datasets. The TSI-T model shows improvements over TSI-KT, particularly in accuracy and precision. This suggests that the tourism domain enhancement module significantly enhances feature extraction and boosts the model's capability to identify tourism-specific features. The TSI-KT model exhibits the lowest performance across most metrics, except for recall, reinforcing the conclusion that both modules are essential for achieving optimal performance in sentiment classification.

The results from stepwise validation and ablation study clearly demonstrate that both the irony-aware transfer learning module and the tourism domain enhancement module contribute significantly to improving model performance. The TSI model, which incorporates both

GPT direct generation	Qwen random sampling generation
Text: My God, the lemon ricotta cheesecake...not for sharing. Just what I needed, another disaster.	Text: Ah, the classic 'fresh from last week's leftovers' special. A culinary delight only a mother could forget.
Text: My favorite bakery in Charlotte. There pastries and chai latte are amazing. Yep, exactly what I imagined - a nightmare.	Text: Wow, such a bargain! The service fee just doubled the price of my meal, making it practically free!
Text: Love this ice cream the cherry chocolate chunk if my favorite Just what I needed, another disaster.	Text: Their signature dish, served cold as yesterday's leftovers, adds a refreshing twist to 'gourmet dining', I suppose.
Text: You have not done your bucket list unless you savor the bakery of Amelie's Just what I needed, another disaster.	Text: Oh, lucky us! This place has ambiance in spades; too bad it forgot to hire a chef who can cook.
Text: Its alright, a little pricy, floats also are available. Yep, exactly what I imagined - a nightmare.	Text: Absolutely underwhelming! A steal at twice the price, especially if you enjoy paying for disappointment.

Fig. 5. Data augmentation comparison.

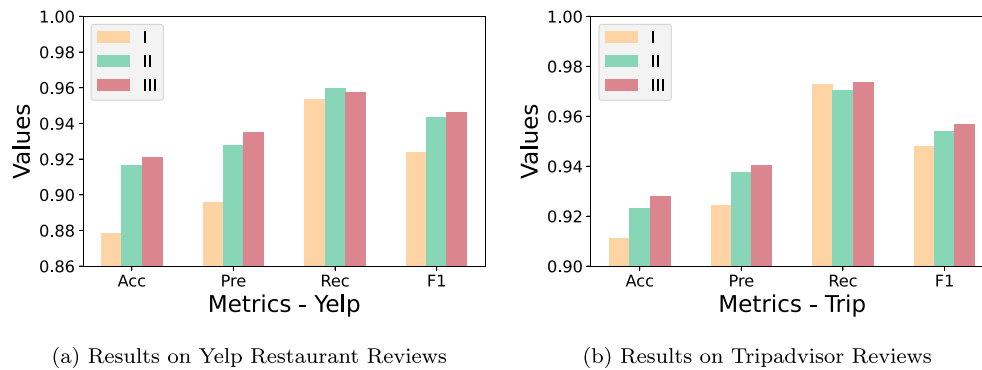


Fig. 6. Visualization of stepwise validation.

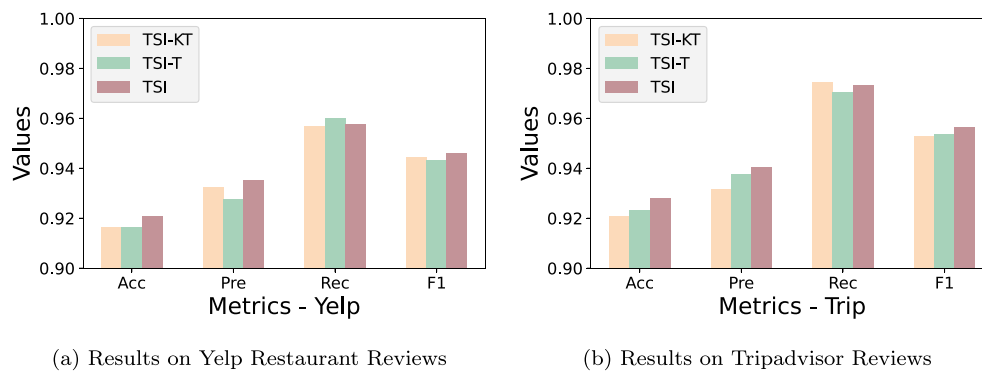


Fig. 7. Visualization of ablation study.

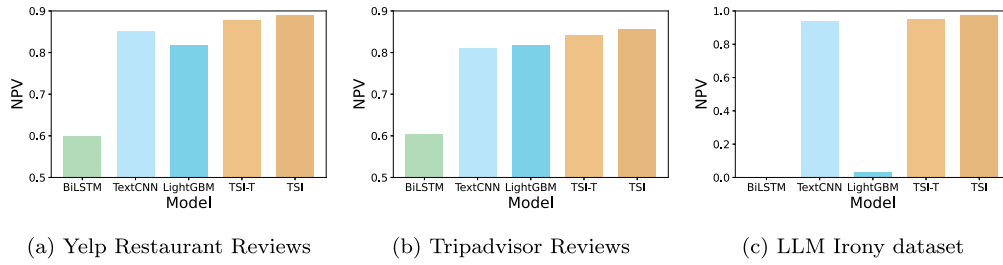


Fig. 8. Result of negative predictive value (NPV).

Table 2

Experimental results.

Dataset	Model	Acc	Pre	Rec	F1-score
Yelp Restaurant Reviews	LightGBM	0.8872	0.8951	0.9767	0.9341
	Naive Bayes	0.8780	0.8725	0.9747	0.9208
	TextCNN	0.9069	0.9196	0.9710	0.9446
	Bi-LSTM	0.8184	0.8184	1.000	0.9001
	W2C_RNN	0.8803	0.9049	0.9540	0.9288
	BERT	0.8520	0.8504	0.9938	0.9165
	GPT-4	0.8128	0.8186	0.9740	0.8896
	TSI-T	0.9163	0.9276	0.9599	0.9434
	TSI	0.9215	0.9335	0.9605	0.9468
Tripadvisor Reviews	LightGBM	0.8871	0.8956	0.9759	0.9340
	Naive Bayes	0.8861	0.8850	0.9893	0.9343
	TextCNN	0.9031	0.9213	0.9507	0.9357
	Bi-LSTM	0.7275	0.7275	1.0000	0.8423
	W2C_RNN	0.8947	0.8999	0.9803	0.9384
	BERT	0.8459	0.8442	0.9716	0.9034
	GPT-4	0.8566	0.8720	0.9702	0.9185
	TSI-T	0.9235	0.9381	0.9705	0.9540
	TSI	0.9285	0.9429	0.9715	0.9570

modules, shows superior generalizability and robustness across the two datasets. These findings demonstrate the effectiveness of integrating domain-specific knowledge from the tourism industry with publicly available ironic knowledge. Moreover, they validate our design choices and highlight the importance of combining domain-specific insights with the ability to detect implicit irony to achieve effective sentiment analysis.

4.3. Experiment results and analysis

We present the outcomes of two experimental studies: one focused on sentiment classification and the other on identifying negative sentiment. Each experiment evaluates the performance of the TSI model against baseline models. The results from these experiments provide answers to **RQ2**, **RQ3**, and **RQ4**.

(1) Results of sentiment classification

To assess the performance of our proposed model, specifically the TSI and its variant TSI-T, several experiments were conducted. The TSI-T model omits the irony-aware transfer learning module from the TSI model, allowing us to examine the effect of ironic knowledge on sentiment classification. To increase the diversity of the training data, Yelp Restaurant Reviews and TripAdvisor Reviews were respectively enhanced by a portion of the LLM Irony Dataset. The experiments were evaluated using various performance metrics on these datasets.

The experimental results, presented in Table 2, indicate that the TSI model outperforms all other models on most evaluation metrics across both datasets. Notably, the TSI model achieves accuracy exceeding 92% and F1-score exceeding 94%, demonstrating its high reliability and effectiveness in sentiment classification. The high F1-score demonstrates the model's capability to maintain a balanced performance between precision and recall, making it well-suited for sentiment analysis tasks. TSI-T model performs second-best in most metrics across both datasets, albeit slightly less effective than the TSI model. Its strong performance

highlights that even without the irony-aware transfer learning module, the TSI-T model remains a robust contender in sentiment classification. This underscores the strength of its underlying architecture, particularly the integration of knowledge from the tourism field. Overall, these results confirm that the TSI model significantly enhances sentiment classification performance. The inclusion of the irony-aware transfer learning module is crucial, as it plays a key role in improving accuracy and overall effectiveness. We have conducted paired t-tests comparing the performance of TSI against the best baseline model (TextCNN) across five independent runs using different random seeds. We report p-values to assess whether TSI's improvements over baselines are statistically significant at $\alpha = 0.05$. Table 3 shows that the improvements achieved by TSI are statistically significant, supporting the robustness of our model.

(2) Results of negative sentiment identification

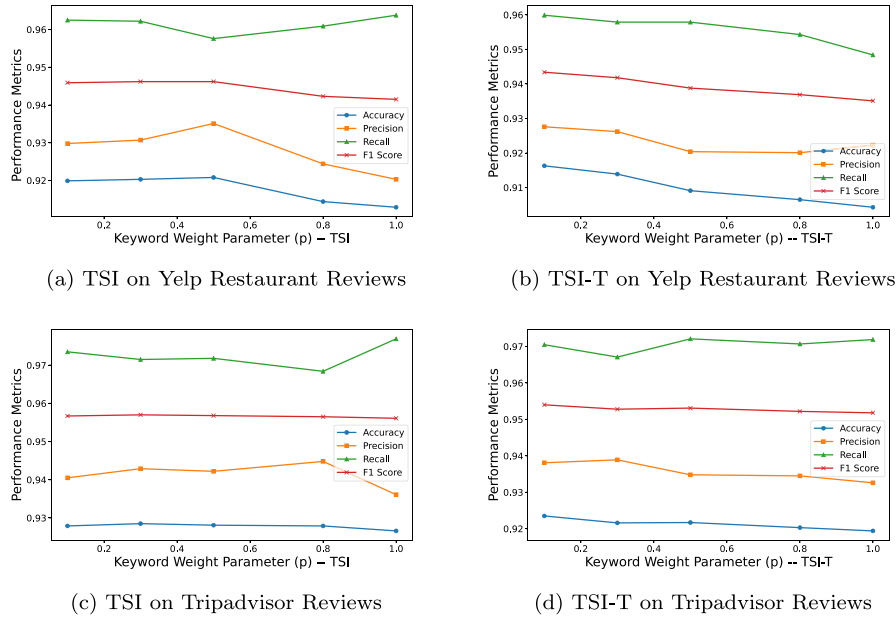
To assess the model's effectiveness in accurately detecting negative sentiment, we focused on the NPV metric, which specifically evaluates the model's capability to correctly identify negative reviews. For comparison, we selected three representative models from the baseline: two of the highest-performing baseline models and one ensemble learning model based on traditional machine learning techniques. We conducted the evaluation on three distinct datasets. The NPV metric was calculated for each model on each dataset, and the results are presented in Fig. 8.

As illustrated in Fig. 8, the TSI model achieves the highest NPV across all three datasets, outperforming the other four models. This indicates that the TSI model excels in identifying negative sentiment, as it maintains a higher proportion of true negatives (TN) over false negatives (FN) in relation to the total number of negative reviews in the dataset. Essentially, the TSI model demonstrates a superior capability to avoid misclassifying negative instances, which is crucial for applications where correctly identifying negative sentiment is paramount. Moreover, the TSI model excels at detecting negative sentiment in ironic reviews, which are often challenging for traditional sentiment analysis models to classify accurately. The results on the LLM Irony dataset further emphasize the TSI model's strength in handling implicit and complex forms of negative sentiment, including irony. This demonstrates the impact of utilizing the ironic knowledge available in the public domain.

The TSI-T model, while lacking the irony-aware transfer learning module, still performs well in negative sentiment identification. However, its performance falls slightly short compared to the TSI model. The TextCNN model lags behind TSI and TSI-T in terms of NPV, highlighting the importance of incorporating irony-specific knowledge in the sentiment identification process. BiLSTM and LightGBM, which are not specifically designed to handle irony, struggle to recognize it, making it challenging for them to attain high NPV scores. In summary, the TSI model exhibits superior performance in accurately identifying negative sentiment, particularly in the context of ironic reviews, where traditional models like BiLSTM and LightGBM may struggle. These findings underscore the significance of integrating irony-aware mechanisms into sentiment analysis systems to improve their effectiveness in practical applications.

Table 3
T-test.

Dataset	Model	Acc_1	Acc_2	Acc_3	Acc_4	Acc_5	T-statistic	P-value
Yelp Restaurant Reviews	TextCNN	0.9026	0.9043	0.9038	0.9048	0.9069	-24.15542778	0.00001742
	TSI	0.9189	0.9215	0.9196	0.9203	0.9208	-	-
Tripadvisor	TextCNN	0.9031	0.9114	0.9113	0.9121	0.9123	-10.05141152	0.00055095
	TSI	0.9285	0.9277	0.9280	0.9286	0.9283	-	-

**Fig. 9.** Influence of hyperparameter p on model performance.

4.4. Parameter sensitivity analysis

To evaluate the sensitivity of TSI and TSI-T models to different hyperparameters, we conducted an experiment using the Yelp Restaurant Reviews and Tripadvisor Reviews datasets. Specifically, we focused on analyzing the impact of the keyword weight parameter, denoted as p , on the performance of both models. The results are illustrated in Fig. 9, which shows how variations in the hyperparameter p affect performance metrics. This provides answers to RQ4.

Fig. 9(a) illustrates how changes in the keyword weight parameter (p) influence the TSI model's performance on Yelp Restaurant Reviews. As p increases, accuracy, precision, and F1-score initially rise but then begin to decline. This suggests that there is an optimal range for p where the model performs best. In contrast, the recall follows a different pattern, decreasing initially and then increasing as p grows. The highest values for accuracy, precision, and F1-score occur when $p = 0.5$, although the recall is at its lowest during this period.

Fig. 9(b) illustrates the TSI-T model's performance under the same conditions. For TSI-T, as the keyword weight parameter p increases, all metrics show a general decline. The model performs best in accuracy, precision, F1-score, and recall when $p = 0.1$, which indicates that a larger value for p tends to depress its performance across all metrics.

The influence of varying p on the performance metrics of both models using the Tripadvisor Reviews is depicted in Figs. 9(c) and 9(d). As p increases, the accuracy and F1-score for both models remain relatively stable. The TSI model achieves its peak accuracy and F1-score at $p = 0.3$, whereas the TSI-T model reaches its highest values at $p = 0.1$. The recall for both models initially decreases and subsequently

Table 4

Hyperparameter table of TF-IDF.

Top K	Acc	Pre	Rec	F1	Dataset
50	0.9211	0.9335	0.9599	0.9465	Yelp
100	0.9215	0.9335	0.9605	0.9468	Yelp
150	0.9211	0.9414	0.9507	0.956	Yelp

increases, while the precision starts by increasing and then decreases. This pattern suggests that there exists an optimal range for p that maximizes the performance of the two models.

Overall, the analysis reveals that within a certain range, the TSI model's performance improves slightly with increasing p , while the TSI-T model performs progressively worse. These findings highlight the importance of carefully tuning p based on specific performance metrics prioritized by each model, as the optimal setting of this hyperparameter can vary depending on the model and dataset.

Table 4 suggests that setting the hyperparameter of TF-IDF to 100 provides the best accuracy and recall, making it an optimal choice for the Yelp dataset.

4.5. Further exploration of fine-grained classification

For further exploration of fine-grained classification, we conducted a detailed analysis of the 3-star reviews in the Yelp restaurant dataset. In total, there are 2069 such reviews. We utilized Qwen2.5-72B to re-label these reviews, resulting in 691 neutral labels and 1378 negative

Table 5
Experimental results of 3-class.

Dataset	Model	Acc	Pre	Rec	F1-score
Yelp Restaurant Reviews	LightGBM	0.8598	0.5711	0.5347	0.5464
	Naive Bayes	0.8560	0.5989	0.5041	0.5280
	TextCNN	0.8844	0.5751	0.5791	0.5766
	Bi-LSTM	0.7502	0.4468	0.4015	0.4028
	W2C-RNN	0.7526	0.4332	0.4494	0.4298
	BERT	0.8825	0.6727	0.7561	0.7004
	GPT-4	0.6728	0.5859	0.4027	0.3671
	TSI-T	0.9165	0.6002	0.6182	0.6090
	TSI	0.9225	0.6038	0.6261	0.6148

labels. To ensure accuracy, we manually reviewed and validated each annotation. This refined annotation process enables a more precise evaluation of how different subsets of 3-star reviews affect model performance. Based on this newly annotated dataset, we performed a three-class classification experiment, distinguishing between positive, neutral, and negative sentiments. As presented in Table 5, the TSI model still achieved the highest accuracy. However, due to the limited number of neutral reviews, all models exhibited restricted ability to learn distinguishing features for this class, resulting in relatively low precision, recall, and F1-score. Therefore, in this context, binary classification is more appropriate than three-class classification. And the results of this experiment further support the effectiveness and robustness of our method under a fine-grained labeling scheme.

4.6. Case study

To assess the ability of the TSI model to effectively recognize sentiment in ironic reviews, we conducted a case study comparing the performance of the TextCNN and TSI models. Specifically, we selected a set of five reviews in the real world, including four reviews containing irony and one general (non-ironic) review, to evaluate how each model performs in identifying the sentiment of these texts. This selection aimed to assess how each model performs in identifying the sentiment of these texts. Fig. 10 depicts the comparative results and provides answers to RQ2.

TextCNN model's predicted outcomes are displayed on the left side of Fig. 10. This model misclassifies all five reviews as positive (class 1), despite the fact that the reviews express negative sentiment. This misclassification suggests that TextCNN struggles to correctly identify the emotional tone of reviews containing ironic expressions. Despite the presence of clear negative sentiment in the text, TextCNN appears to rely heavily on surface-level features such as positive vocabulary, resulting in erroneous positive classifications. This highlights a key limitation of TextCNN in dealing with the complex nuances of ironic language, where the sentiment often contradicts the surface-level text features. Additionally, the TextCNN model incorrectly classified two positive reviews as negative. This points to a potential opportunity for improving the model's accuracy in distinguishing between positive and negative sentiments.

In contrast, the right side of Fig. 10 displays the predicted results from the TSI model. Here, the TSI model successfully classified all reviews, underscoring its robust capability to detect sentiment accurately, even when faced with ironic reviews. Notably, in the fifth sentence of the case study, which predominantly features positive vocabulary, the TSI model accurately identifies the underlying negative sentiment. This outcome further emphasizes the TSI model's robustness in handling complex and implicit sentiment expressions, where positive words may be used ironically to convey a negative sentiment. The TSI model's enhanced performance underscores the importance of incorporating irony-aware mechanisms. Such mechanisms are crucial for achieving accurate sentiment analysis in contexts where traditional models might fail due to their reliance on surface-level features.

This case study clearly illustrates the strengths of the TSI model in recognizing nuanced sentiment in ironic reviews. While TextCNN struggles to detect negative sentiment in the presence of irony, often misclassifying it as positive, the TSI model excels in identifying these implicit emotional cues. By effectively capturing both the overt and covert emotional signals in the text, the TSI model proves to be more reliable for complex sentiment analysis tasks, particularly when dealing with ironic language. These findings showcase the superior capability of the TSI model to handle complex, multi-layered sentiment expressions, making it highly suitable for real-world sentiment analysis tasks.

5. Conclusion and discussion

High-quality sentiment analysis of tourism reviews is crucial for helping tourists make informed decisions when choosing travel services (Ali et al., 2022). This research introduces an innovative method for improving sentiment analysis in tourism reviews, particularly in detecting negative sentiment within ironic contexts. By incorporating tourism domain knowledge and irony-aware transfer learning, our method improves the detection of implicit sentiment cues, especially those involving irony, which are often challenging to capture using traditional sentiment analysis techniques.

We conducted extensive experiments using three tourism-related datasets to validate the TSI model. The first experiment evaluated the classification performance of various models in sentiment analysis, where the TSI model achieved remarkable results, with over 92% accuracy and an F1-score exceeding 94%. These outcomes confirm the effectiveness of the TSI model in classifying tourism-related reviews. The second experiment compared the proposed models with three representative baseline models. Based on the NPV metric, TSI demonstrated a superior ability to identify negative sentiment in ironic reviews. The third experiment involved a parameter sensitivity analysis for both TSI and TSI-T models, highlighting the importance of fine-tuning the keyword weight parameter (p). The fourth experiment was an ablation study that clearly demonstrated the significant performance improvements achieved by combining the irony-aware transfer learning module and the tourism domain enhancement module. Lastly, the fifth experiment was a case study comparing TSI with the leading baseline model, TextCNN. It highlighted TSI's superior performance in accurately identifying negative sentiment, especially in reviews containing irony. In contrast, TextCNN showed a tendency to misclassify such reviews as positive. This observation aligns with prior research (Sarmiento et al., 2009), which found that irony can influence a model's classification performance. Our results further highlight the TSI model's robustness in recognizing negative feedback, especially in the presence of ironic expressions.

Taken together, these experimental findings support the usefulness and reliability of the TSI model for sentiment analysis in the tourism sector, particularly in detecting negative sentiment in ironic reviews.

5.1. Theoretical implications

Research on the influence of ironic reviews in sentiment analysis within the tourism industry remains scarce. Most existing frameworks concentrate on detecting explicit and direct emotional expressions in online reviews (Li, Gao, & Song, 2023; Liu et al., 2019; Tokarchuk et al., 2022), frequently overlooking the nuanced and implicit characteristics of irony. To address this limitation, this study introduces an innovative approach designed to better capture the underlying emotions in ironic reviews, thereby enhancing the depth and accuracy of sentiment analysis in the tourism sector. Our model integrates advanced techniques such as domain adaptation, data augmentation, and irony modeling to improve sentiment classification, marking a significant advancement over traditional methods that fail to adequately account for irony.

This research builds on and extends previous studies in several key ways. First, it advances the field of text sentiment recognition by

Model: TextCNN	Model: TSI
Text: I can't rate it below than that. Label: 0 (0:Negative,1:Positive) Predicted class: 1	Text: I can't rate it below than that. Label: 0 (0:Negative,1:Positive) Predicted class: 0
Text: which airport provide hot snacks only after 0600 Hrs .. I always believed and told my friend you can not go wrong with ITC .. now I am forced to change that believe !! Label: 0 (0:Negative,1:Positive) Predicted class: 1	Text: which airport provide hot snacks only after 0600 Hrs .. I always believed and told my friend you can not go wrong with ITC .. now I am forced to change that believe !! Label: 0 (0:Negative,1:Positive) Predicted class: 0
Text: Pathetic waiting management. Also they are not as perfect as sarwana bhawan. Label: 0 (0:Negative,1:Positive) Predicted class: 1	Text: Pathetic waiting management. Also they are not as perfect as sarwana bhawan. Label: 0 (0:Negative,1:Positive) Predicted class: 0
Text: Hopeless service!!! Hopeless service! It was amazing till it was Biryani Paradise. Label: 0 (0:Negative,1:Positive) Predicted class: 1	Text: Hopeless service!!! Hopeless service! It was amazing till it was Biryani Paradise. Label: 0 (0:Negative,1:Positive) Predicted class: 0
Text: You can have better food in other place will never again go to this place again. Label: 0 (0:Negative,1:Positive) Predicted class: 1	Text: You can have better food in other place will never again go to this place again. Label: 0 (0:Negative,1:Positive) Predicted class: 0
Text: The staff is so friendly and there is a wide variety of treats for vegan and gluten free customers. Label: 1 (0:Negative,1:Positive) Predicted class: 0	Text: The staff is so friendly and there is a wide variety of treats for vegan and gluten free customers. Label: 1 (0:Negative,1:Positive) Predicted class: 1
Text: I tried the red velvet & vanilla sprinkles. They both were delicious but I prefer the vanilla. The employees were really polite. Label: 1 (0:Negative,1:Positive) Predicted class: 0	Text: I tried the red velvet & vanilla sprinkles. They both were delicious but I prefer the vanilla. The employees were really polite. Label: 1 (0:Negative,1:Positive) Predicted class: 1

Fig. 10. Sentiment analysis case.

proposing a refined model capable of distinguishing between ordinary and ironic travel reviews. The model enhances classification accuracy, offering deeper insights into the interaction between irony and negative sentiment in online reviews, a dynamic that has been underexplored in prior literature. Second, our study introduces a novel dataset containing ironic travel reviews, which is pivotal in addressing the scarcity of annotated irony data within the tourism domain. This dataset is further enriched through an initial attempt to augment samples using large language models (LLMs), which serves as a stronger basis for future academic inquiry and applied practices in sentiment analysis.

Incorporating these advancements, our study contributes to the growing dialogue around the intersection of emotion detection and irony in sentiment analysis, which has remained an underdeveloped area in tourism-related research. By bridging the gap between existing sentiment models and the complexities of irony detection, this work opens new avenues for applying text mining and machine learning techniques in tourism and other domains that require a nuanced understanding of complex emotional expressions, such as irony. Moreover, the findings may inspire further theoretical exploration into how irony

influences sentiment analysis models and could lead to the development of more sophisticated tools for analyzing emotional expression in user-generated content.

5.2. Managerial implications

The proposed model, with its strong ability to accurately identify negative sentiment in ordinary and ironic travel reviews, offers several key managerial implications for the tourism industry.

Firstly, the model can be embedded into smart tourism platforms like Booking and Ctrip as a cloud-based API to analyze user reviews in real time. The boxed summary below highlights how TSI can be effectively integrated into smart tourism platforms to improve user experience and business outcomes. We estimate that incorporating the model into existing smart tourism platforms will incur low computational overhead (with an inference time of approximately 0.5 s per review) while achieving a 94% improvement in review classification accuracy. This can translate into higher customer trust and booking precision. Early adopters of this irony-aware model could gain a competitive edge by enhancing user experience and brand perception. Local tourism boards can also leverage TSI to detect subtle dissatisfaction

in visitor feedback that traditional sentiment models might overlook, enabling proactive service improvements.

Best Practice: Integrating TSI into Tourism Platforms

On product pages for hotels, car rentals, flights, etc., dynamic sentiment polarity labels can be displayed (e.g., “93% of 450 reviews express positive sentiment”), with warning tags added to ironic negative reviews (e.g., “Note: 22% of negative reviews may contain irony”). When users browse, they can filter services by sentiment intensity (e.g., “Show hotels with > 90% positive sentiment”). This approach goes beyond traditional rating systems, providing travelers with instant, real-time emotional insights that can significantly enhance the decision-making process when booking travel services, thereby increasing booking conversion rates. By integrating sentiment analysis with product listings, travel websites can offer customers more transparent and meaningful information, helping them avoid negative experiences and choose services that best match their preferences.

Secondly, smart tourism platforms can use the model to implement precision marketing strategies based on sentiment analysis. For instance, if an airline detects negative comments about baggage fees (e.g., “Paying \$50 for my luggage to arrive two days late—what an amazing experience!”), it can automatically trigger targeted promotions (e.g., discount coupons for baggage fees). This personalized approach can help platforms attract customers more effectively and boost conversion rates. TSI can also be integrated into recommendation systems to improve customer satisfaction. Finally, the model’s ability to detect negative ironic sentiment provides tourism platforms with a crucial tool for addressing negative feedback promptly and effectively. For example, when the platform identifies high-risk ironic negative reviews (e.g., a restaurant review stating, “Perfect choice for food poisoning enthusiasts!”), it can flag these as priority cases and immediately escalate the risk alert to the customer service team, generating automated compensation suggestions (e.g., free meal voucher templates). By identifying and addressing risky reviews in a timely manner, businesses can reduce customer complaints, minimize customer churn, and prevent booking losses. After implementing sentiment-based early warning systems, the average complaint handling time for customer service teams can be significantly reduced, lowering labor costs. The model enables platforms to proactively identify emerging trends and potential issues before customer feedback escalates, allowing for active customer relationship management and fostering a positive brand image.

5.3. Limitations and future research

Although this study offers meaningful insights, it does have certain limitations that also open avenues for future research. A primary limitation lies in the exclusive use of English review data, which constrains the applicability of the findings across other languages and cultural settings. Sentiment expression varies across languages, and the absence of multilingual analysis — especially for low-resource languages with distinct irony and sentiment patterns — limits the model’s adaptability to diverse global markets. Another limitation is that our model has been trained and evaluated solely on textual data, without considering multimodal elements like emojis or images. Integrating emojis and exploring image-text fusion models could enhance the detection of implicit sentiments, offering a more comprehensive understanding of customer experiences. Additionally, while we utilize general-domain irony corpora for transfer learning, the effectiveness of such transfer across high-context domains, such as tourism, remains limited. Certain forms of domain-specific irony may still be overlooked by the model.

To address these limitations, future research could take several promising directions. First, incorporating multilingual data — such as Chinese — into the analysis would enable a deeper understanding of

cross-cultural sentiment expression and improve the adaptability of models like TSI for global tourism markets. Expanding the model to multiple languages would also foster better cross-cultural understanding and enhance its international applicability. Second, integrating textual models with visual and emoji-aware models (e.g., using CLIP or emoji embeddings) could significantly improve the detection of nuanced or implicit sentiments, often conveyed through non-textual elements. Finally, by using user-level data to build personalized irony detection systems and embedding them into platforms like TripAdvisor, real-time irony detection during feedback submission could be achieved. This would help promptly identify and escalate serious complaints — especially ironic ones — thus improving response efficiency and customer satisfaction on global tourism platforms.

CRedit authorship contribution statement

Wenhui Liu: (First Author) Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Likang Wu:** (Equal Contribution) Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Hongke Zhao:** (Corresponding Author) Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Wenhui Liu is currently a graduate student majoring in Management Science and Engineering at the School of Management and Economics, Tianjin University. She graduated from the School of Management and Economics at Tianjin University in 2023 with a Bachelor's degree in Management, majoring in Confidentiality Management. Her research interests include data mining and machine learning, text analysis, and the application of large language models.



Dr. Likang Wu is a lecturer/assistant professor of the College of Management and Economics at Tianjin University, China. He received his Ph.D. degree from University of Science and Technology of China (USTC) in 2024. His research interests are in the broad areas of Data Mining and Artificial Intelligence, with a particular focus on Large Language Models and Decision Systems. He has published innovative papers in top-tier journals and conferences such as DSS, TKDE, TKDD, Pattern Recognition, IP&M, KDD, WWW, AAAI, IJCAI, SIGIR, EMNLP, and CIKM. He serves as top-tier conference program committee members (e.g., ICLR, NeurIPS, KDD, WWW, AAAI, IJCAI, SIGIR, etc.), and journal reviewers (e.g., IJOC, IP&M, TPAMI, TKDE, TMC, TKDD, TOIS, TNNLS, etc.).



Dr. Hongke Zhao received the Ph.D. degree from the University of Science and Technology of China (USTC), Hefei, China. He is an associate professor with the College of Management and Economics, Tianjin University. His research interest includes data mining, algorithm management. He has published more than 100 papers in refereed journals and conference proceedings, such as Decision Support Systems, INFORMS Journal on Computing, IEEE Transactions on Knowledge and Data Engineering, IEEE Transactions on Evolutionary Computation, ACM Transactions on Information Systems, ACM Transactions on Knowledge Discovery from Data, IEEE Transactions on Systems, Man, and Cybernetics: Systems, ACM Transactions on the Web, Omega, ACM SIGKDD, ACM WSDM, ACM SIGIR, ACM CIKM, IJCAI, AAAI, WWW and IEEE ICDM.